CIM-Pruner: A Dual-Mode Compute-In-Memory Macro for Efficient VLMs with Intra-Chunk Token Pruning and Merging

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***Abstract*—Vision-language models (VLMs) have demonstrated remarkable capabilities under various downstream tasks but incur significant computational overhead, especially during edge deployment. While text-guided vision token pruning algorithm effectively reduces redundancy, existing hardware architectures lack tailored support for such dynamic patterns, especially for token-level pruning and merging. We propose CIM-Pruner, a hardware-algorithm co-design framework for efficient VLM inference on digital compute-in-memory architectures. CIM-Pruner introduces: (1) an intra-chunk token pruning and merging algorithm that eliminates expensive global top-k sorting by intra-chunk incomplete sorting, (2) an early-quit bitonic sorting circuit for sorting and labeling different input tokens with reasonable area and power consumption, and (3) a dual-mode token processing unit featuring a dual-mode Booth multiplier and accumulator for differentiated token processing. Implemented in 28nm CMOS, evaluation results show that the proposed CIM-Pruner macro achieves an energy efficiency of 25.64 TOPS/W (@200MHz) featuring INT8 data precision.**

***Index Terms*—Computing-in-memory (CIM), vision-language models, text-guided token pruning, token merging.**

I. Introduction

Vision-language models(VLMs), such as LLaVA [1], have emerged as foundational architectures for AI applications across diverse scenarios, demonstrating unprecedented capabilities across various tasks. However, deploying VLMs on resource-constrained edge devices remains a formidable challenge. A large number of visual tokens results in excessive memory access and energy consumption, far exceeding the capabilities of edge processors.

Digital compute-in-Memory (CIM) architectures eliminate intensive data communication by integrating computing logic into memory, achieving remarkable energy efficiency improvements. However, existing text-guided token pruning methods, particularly top-k sorting-based approaches such as Sparse-VLMs [2], are fundamentally ill-suited for CIM architectures due to their substantial overhead in candidate token selection and lack of hardware support for token merging operations, rendering efficient sparsity exploitation in VLM inference a pressing challenge.

Therefore, the hardware–algorithm mismatch gives rise to two critical challenges that severely limit the efficiency of VLM inference on CIM architectures. The first challenge is that the global top-k token selection process, such as heap-based selection, which is fundamental for identifying important vision tokens in many pruning algorithms, requires global sorting operations with high computational complexity. Digital CIM architectures, although promising for their energy efficiency and highly localized parallel processing, are inherently incompatible with such global sorting and selection procedures as Fig. 1(b) illustrates. The second challenge is that to avoid the permanent loss of potentially useful information, modern token pruning algorithms often adopt token merging strategies that average and combine less important tokens into new ones, just as Fig. 1(c) shows. However, current CIM architectures, designed for dense and uniform matrix operations, lack intrinsic support for differentiated token processing such as token merging.

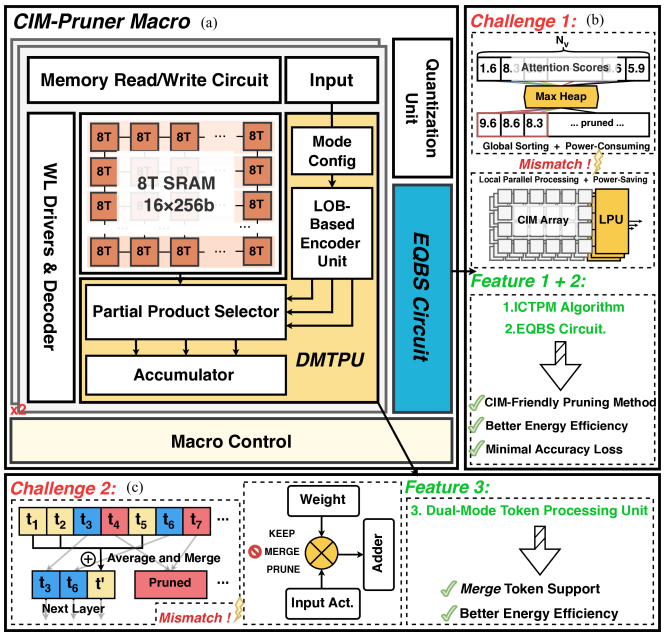


Fig. 1. (a) Overall architecture of the proposed CIM-Pruner macro. (b) Challenge 1. (c) Challenge 2.

To address these challenges, we present CIM-Pruner, a hardware-algorithm co-design framework that bridges the gap between text-guided vision token pruning algorithms and digital CIM hardware through three levels of innovation. The main contributions of this work are as follows:

- We introduce an intra-chunk token pruning and merging algorithm that eliminates the need for expensive global sorting by partitioning vision tokens into fixed-size

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chunks and assigning differentiated pruning or merging strategies based on chunk-wise importance metrics.
- We design an early-quit bitonic sorting circuit that supports incomplete sorting and quartile-based labeling with minimal area overhead and power consumption.
- We propose a dual-mode token processing unit, providing enhanced support for token pruning and merging with remarkable reduction in power consumption.

The overall architecture is shown in Fig. 1(a). For validation, an INT8 CIM-Pruner macro is implemented in 28nm CMOS technology and achieves an energy efficiency of 25.64 TOPS/W (@0.9V, 200MHz).

II. Background and Related Work

A. Vision Token Pruning and Token Merging

VLMs integrate computer vision and natural language processing by tokenizing visual inputs into patch embeddings, processing them through a Vision Transformer encoder and projector, and subsequently concatenating with text tokens for self-attention computation.

Previous works on vision token pruning fall into two paradigms, depending on whether textual tokens are used. *Text-agnostic methods* like VisionZIP [3] and FasterVLM [4], rely only on visual saliency and ignore textual modality, thus performing poorly in certain scenarios. In contrast, *text-guided methods* incorporate textual semantics via cross-modal attention between text 'rater' tokens and vision tokens. SparseVLMs pioneered this approach, retaining high-attention tokens relevant to the query while pruning low-scoring ones.

Besides, to prevent important information from being permanently deleted, recent algorithms introduce the *token merging* strategy that consolidates similar tokens with relatively high scores rather than discarding them entirely. ToMe [5] first introduced token merging to reduce the token count while maintaining a complete representation of the visual input, which inspired the subsequent token pruning works.

B. Digital Compute-in-Memory for Transformers

Digital CIM architectures have emerged as a promising solution for accelerating Transformers. Earlier studies such as TranCIM [6] and RedCIM [7] have laid the groundwork for CIM-based accelerators targeting transformer models. However, conventional CIM architectures still provide limited support for sparsity-aware computation and token pruning. To address this limitation, subsequent research has developed sparsity-aware CIM accelerators by reorganizing dataflows, optimizing memory access patterns, and refining computation scheduling. SPCIM [8] focuses on weight reordering to address the under-utilization of the multi-macro architecture, while SUN [9] introduces a reconfigurable shift-and-adder design along with structured pruning support. However, when it comes to token pruning and token merging, most existing CIM accelerators still lack adequate architectural support.

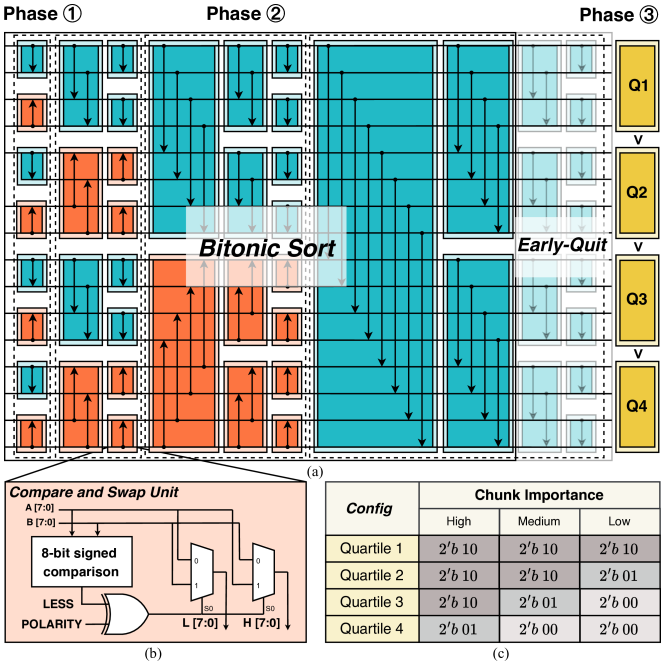


Fig. 2. (a) Architecture of the EQBS. (b) The Compare-and-Swap unit in the EQBS. (c) Scheme of labeling the quartile within a chunk.

III. Proposed CIM-Pruner

A. Intra-Chunk Token Pruning and Merging Algorithm

The current text-guided vision token pruning algorithm uses global top-k selection, which is incompatible with the CIM architecture. Therefore, we present an intra-chunk token pruning and merging (ICTPM) algorithm, which partitions tokens into consecutive chunks of N_c tokens each and performs intra-chunk token selection to better align with the CIM architecture for parallelism.

To be specific, we set N_c = 16 to achieve a balance between accuracy and hardware efficiency. Each chunk C_j contains 16 tokens $\{v_{16j}, v_{16j+1}, \ldots, v_{16j+15}\}$, where j is the index of the chunk. For token v_i, the importance score is the mean cross-modal attention from rater tokens:

$$s_i = \frac{1}{|R|} \sum_{r \in R} \text{Attn}(r, v_i) \tag{1}$$

where R is the rater tokens set.

To accommodate different pruning ratios and better align with the hardware architecture, chunks are ranked and divided into three categories. Each token within a chunk is assigned a 2-bit label, with *KEEP* represented by 10, *MERGE* by 01, and *PRUNE* by 00. Depending on the chunk importance, the ratios of KEEP, MERGE, and PRUNE tokens are adjusted accordingly. In high-importance chunks, 12 tokens are *KEEP*, 4 are *MERGE*; In medium-importance chunks 8 are *KEEP*, 4 are *MERGE*, 4 are *PRUNE*; In low-importance chunks 4 are *KEEP*, 4 are *MERGE*, 8 are *PRUNE*. Every 4 *MERGE* tokens will be averaged and merged to a new token in the further process.

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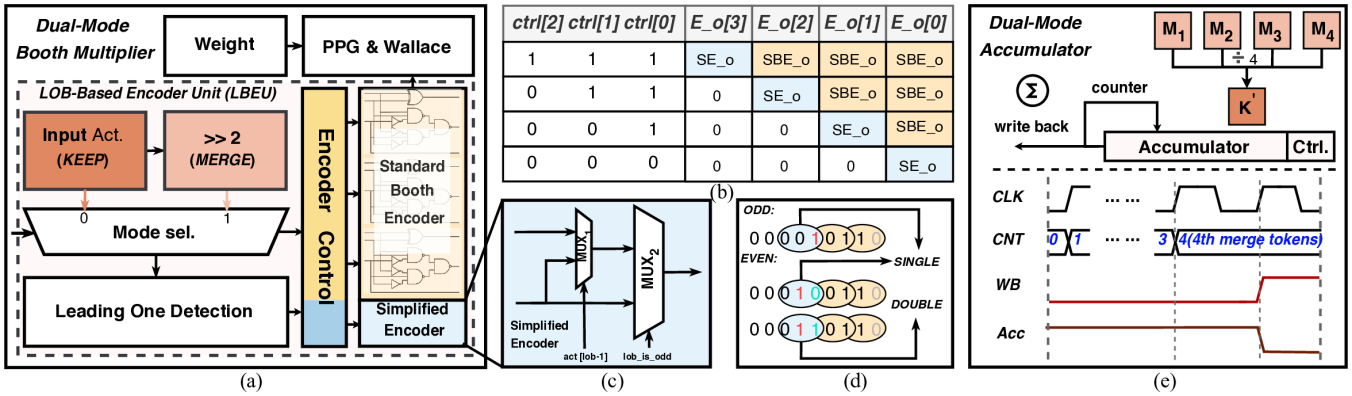


Fig. 3. (a) Overall view of Dual-mode booth multiplier featuring LBEU. (b) Scheme of the LBEU control logic and output generation, SE means Simplified Encoder, and SBE means Standard Booth Encoder. (c) Design of the Simplified Encoder. (d) Details of how the Simplified Encoder works in different conditions. (e) Dual-mode accumulator and corresponding waveform diagram, which illustrates how the *MERGE* mode works.

B. Early-Quit Bitonic Sorting Circuit

The proposed early-quit bitonic sorting circuit (EQBS) is an efficient hardware implementation of the proposed ICTPM algorithm, achieving lower hardware complexity than conventional sorting networks, especially in the CIM architecture. The design rethinks bitonic sorting by noting that there is no need for exact full ranking in ICTPM, which indicates the potential of "early-quit". The general architecture of the EQBS is shown in Fig. 2(a).

The EQBS operates in three pipelined phases. In Phase ①, 16 input tokens with 8-bit importance scores are loaded and partitioned. In Phase ②, tokens are rearranged into a bitonic sequence, and then two comparison stages are applied to sort the whole chunk into four quartiles ($Q_1 > Q_2 > Q_3 > Q_4$), where Q_i represents a set of 4 tokens. This guarantees correct inter-quartile ordering without intra-quartile sorting. In Phase ③, each token is assigned a 2-bit label corresponding to its quartile and the chunk importance through simple logic, which is depicted in Fig. 2(c). For example, within low-importance chunks, $Q_1 \rightarrow 10$, $Q_2 \rightarrow 01$, and Q_3, $Q_4 \rightarrow 00$, which is corresponding to the presented ICTPM algorithm.

The EQBS is built upon a regular and scalable structure, where each compare-swap unit uses an 8-bit comparator and a 2:1 multiplexer, as Fig. 2(b) illustrates, routing max/min values to upper/lower outputs. All comparators at each level operate in parallel, enabling single-cycle-per-stage pipelining.

C. Dual-Mode Token Processing Unit

To gain enhanced support for token pruning and token merging, we design a dual-mode token processing unit (DMTPU) to process tokens with different labels as aforementioned. DMTPU is composed of a dual-mode booth multiplier and accumulator, and the dual-mode booth multiplier features a LOB-based encoder unit, a partial product generator, and a Wallace tree adder.

LOB-based encoder unit (LBEU): According to the presented ICTPM algorithm, 4 consecutive *MERGE* tokens should be averaged and merged to a new token, thus each of them is right shifted by 2 bits before entering the encoder unit, leading to repeated 0 or 1 at the most significant bits. This indicates that some booth encoders are redundant. Therefore, we design the LBEU, including only 3 standard booth encoders and a simplified encoder with extra gate control logic to perform more power-saving computation, especially for *MERGE* tokens, as Fig. 3(a). In detail, ***Firstly***, the input is recognized by labels and then the corresponding form is fed into the leading one detection (LOD) unit, which will output lob (leading one bit) signal that indicates where the first 1 is. ***Secondly***, the encoder control unit will activate only the needed standard encoder and the simplified encoder to save dynamic power, as in Fig. 3(b). To be specific, the number of needed standard encoders N_{se} can be formulated as:

$$N_{se} = \left\lfloor \frac{lob+1}{2} \right\rfloor, \quad N_{se} \leq 3. \tag{2}$$

The number can be reduced because, after the LOD, the encoding logic related to the leading one bit can be replaced by lightweight circuits, which we term a simplified encoder, as Fig. 3(c) and (d) shows. When the lob is odd, the output of the simplified encoder is deterministic; otherwise, it depends on the bit preceding the leading one bit. ***Finally***, the LBEU output together with the 8-bit readout weight will be sent to the partial product generator (PPG) and Wallace tree adder to further complete the computation.

Dual-mode accumulator: As depicted in Fig. 3(e), we redesign the accumulator to offer native support for both *KEEP* and *MERGE* token processing. When set to *MERGE* mode, the Dual-mode accumulator does not write the computation results back until the current 4 *MERGE* tokens are completed. Only minimal additional control logic is needed to keep it accumulating instead of clearing after one token is computed. The accumulator output is expressed as:

$$\mathrm{ACC}_{\mathrm{out}} = \begin{cases} P + \mathrm{ACC}_{\mathrm{prev}}, & \text{if } \textit{KEEP} \\ \sum_{j=0}^{3} P_j + \mathrm{ACC}_{\mathrm{prev}}, & \text{if } \textit{MERGE} \end{cases} \tag{3}$$

where P_j denotes the product for the j-th token among the 4 *MERGE* tokens.

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TABLE I
COMPARISON WITH PRIOR STATE-OF-THE-ART WORKS

	HPCA'20 [10]	TCAS I'23 [11]	JSSC'24 [12]	ISSCC'23 [13]	This Work
Technology	40nm	28nm	28nm	28nm	**28nm**
Data Precision	INT9	INT4/8/16	INT16	INT8	**INT8**
Token Pruning and Merging	No	No	No	No	**Yes**
Supply Voltage (V)	1.1	0.9	0.8	0.66	**0.9**
Frequency (MHz)	1000	200	200	200	**200**
Energy Efficiency (TOPS/W)	1.99 @ INT9	23.24 @ INT8	12.56 @ INT16	18.20 @ INT8[1]	**25.64 @ INT8**[2]

1.Estimated from the voltage scaling figure of energy efficiency. 2.Evaluated in a single Macro with input 25% *KEEP*, 25% *MERGE*, and 50% *PRUNE*.

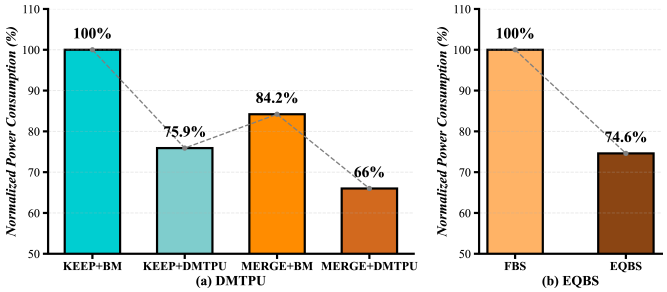


Fig. 4. Energy-saving evaluation of the DMTPU against BM and the EQBS against FBS. (a) For DMTPU, the power consumption is tested in two different modes. (b) For EQBS, the power consumption is compared to FBS.

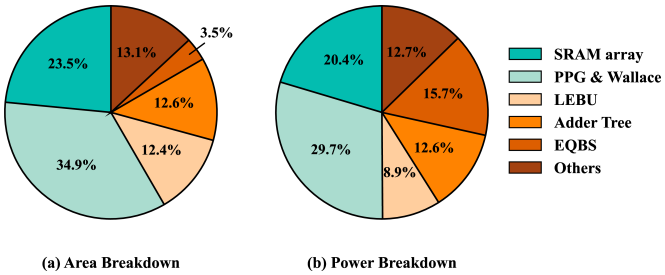


Fig. 5. Area and Power Breakdown of the whole CIM-Pruner macro. (a) illustrates the area breakdown. (b) illustrates the power breakdown.

IV. EVALUATION

A. Experimental Setup

On the algorithm level, we verify the feasibility of the proposed ICTPM algorithm by comparing it with the standard top-k-based pruning method on LLaVA-v1.5-7b. On the hardware level, we implement the macro in 28nm technology, and conduct an ablation study to evaluate the power consumption of the proposed EQBS and DMTPU. Then we assess the area and power breakdown of our entire macro implementation and compare our work with previous state-of-the-art works.

B. Algorithm Validation

The proposed ICTPM algorithm is implemented based on the LLaVA-v1.5-7b model and evaluated on an RTX 4090 GPU. Under the same pruning ratio, we compare ICTPM with SparseVLMs (which adopts standard top-k pruning) on the MME [14], TextVQA [15], and POPE [16] benchmarks. The results show accuracy drops of only 0.92%, 0.75%, and 1.13%, respectively, demonstrating the effectiveness of the proposed ICTPM algorithm.

C. Ablation Study

To gain deeper insight into the power efficiency of the proposed hardware design, we perform an ablation experiment.

First, we evaluate the energy-saving capability of the DMTPU. In this experiment, the SRAM array is configured as 16×64, with dense INT8 weights. The standard booth multiplier (BM) serves as the baseline, with both units operating in *KEEP* and *MERGE* modes. As illustrated in Fig. 4(a), the proposed design achieves an average power reduction of 18.2% in *MERGE* mode and 24.1% in *KEEP* mode, demonstrating its energy efficiency.

We subsequently implement both the EQBS and a conventional full bitonic sorting circuit (FBS) under the same 16 8-bit input configuration as well as some control logic. As shown in Fig. 4(b), EQBS reduces power consumption by 25.4% compared to the FBS, which demenstrates great reduction in power consumption.

D. CIM-Pruner Macro Evaluation and Comparison

Fig. 5 visualizes the evaluated area and power breakdown of the essential parts in the CIM-Pruner macro, which mainly contains the SRAM array, DMTPU, EQBS, and others. Results demonstrate that the SRAM array and the DMTPU are the primary sources of area and power consumption. We then compare our work with prior state-of-the-art counterparts in Table I. Our work achieves 25.64 TOPS/W energy efficiency at INT8 data precision when the input tokens are 25% *KEEP*, 25% *MERGE*, and 50% *PRUNE*.

V. CONCLUSION

This paper introduces CIM-Pruner, a hardware–algorithm co-design framework for efficient VLMs inference on digital compute-in-memory architectures. By incorporating an Intra-Chunk Token Pruning and Merging algorithm, Early-Quit Bitonic Sorting Circuit, and a Dual-Mode Token Processing Unit, CIM-Pruner achieves energy efficiency of 25.64 TOPS/W in a single macro and offers sufficient support for power-saving token pruning and merging.

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